

ADVANCED

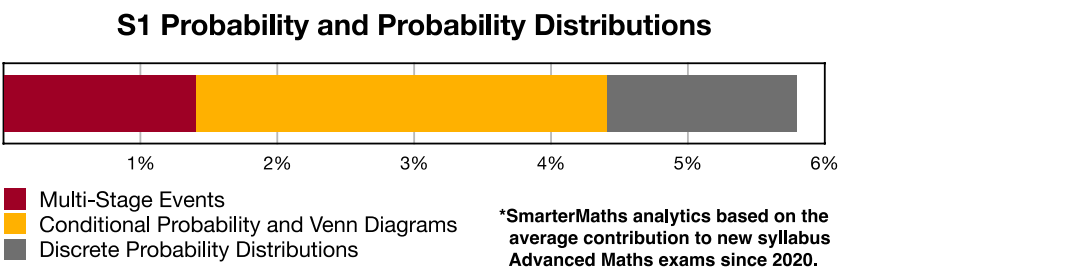
Stage 6

Statistics (Adv), S1 Probability & Probability Distributions (Adv)

Multi-Stage Events (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 135 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- S1 Probability* has contributed an average of 5.8% per new syllabus Advanced exam since it was introduced in 2020.
- S1 Probability* has been split into four sub-categories for the purposes of this analysis: 1-*Multi-Stage Events* (1.4%), 2-*Relative Frequency* (0.0%), 3-*Conditional Probability and Venn Diagrams* (3.0%) and 4-*Discrete Probability Distributions* (1.4%).
- This analysis looks at *Multi-Stage Events*.

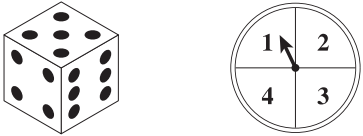
HSC ANALYSIS - What to expect and common pitfalls

- Multi-Stage Events* (1.4%) has been a modest contributor to new syllabus exams as it now has to compete with *Conditional Probability/Venn Diagrams* and *Discrete Probability Distributions*.
- When examined, past HSC cohorts have found *Multi-Stage Events* very challenging with around half the questions producing sub-50% mean marks (2018 was a particularly tough year).
- Tree diagrams can be specifically examined or very useful in answering questions. Note the slightly different tree diagram required in *2008 Adv 7c* and the Marker's Comment.
- Critical to recognise opportunities to apply complementary probability where $P(E) = 1 - P(\text{complement})$. A regularly tested concept last examined in 2024.
- Revision should include examples when the use of an array (table) is efficient (see *2018 Adv 16b*).
- Probability* was combined with the *Series and Sequence* topic in 2013 and 2016, and terribly answered on both occasions!

Questions

1. Probability, 2ADV S1 2023 HSC 2 MC

A game involves throwing a die and spinning a spinner.



The sum of the two numbers obtained is the score.

The table of scores below is partially completed.

		SPINNER			
		1	2	3	4
DIE	1	2	3	4	
	2	3	4	5	
	3		5	6	
	4			7	
	5				
	6				

What is the probability of getting a score of 7 or more?

- A. $\frac{1}{6}$
- B. $\frac{1}{4}$
- C. $\frac{5}{18}$
- D. $\frac{5}{12}$

2. Probability, 2ADV S1 2015 HSC 4 MC

The probability that Mel's soccer team wins this weekend is $\frac{5}{7}$.

The probability that Mel's rugby league team wins this weekend is $\frac{2}{3}$.

What is the probability that neither team wins this weekend?

- A. $\frac{2}{21}$
- B. $\frac{10}{21}$
- C. $\frac{13}{21}$
- D. $\frac{19}{21}$

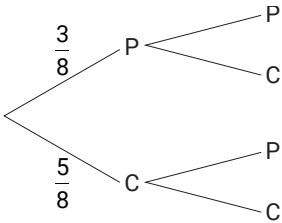
3. Probability, 2ADV S1 2019 MET2-N 9 MC

At the start of a particular week, Kim has three red apples and two green apples. She eats one apple everyday. On Monday, Tuesday and Wednesday of that week, she randomly selects an apple to eat. In this three-day period, the probability that Kim does not eat an apple of the same colour on any two consecutive days is

- A. $\frac{1}{5}$
- B. $\frac{3}{10}$
- C. $\frac{2}{5}$
- D. $\frac{6}{25}$

4. Probability, 2ADV S1 2021 HSC 6 MC

There are 8 chocolates in a box. Three have peppermint centres (P) and five have caramel centres (C). Kim randomly chooses a chocolate from the box and eats it. Sam then randomly chooses and eats one of the remaining chocolates. A partially completed probability tree is shown.



What is the probability that Kim and Sam choose chocolates with different centres?

- A. $\frac{15}{64}$
- B. $\frac{15}{56}$
- C. $\frac{15}{32}$
- D. $\frac{15}{28}$

5. Probability, 2ADV S1 2013 HSC 5 MC

A bag contains 4 red marbles and 6 blue marbles. Three marbles are selected at random without replacement.

What is the probability that at least one of the marbles selected is red?

- A. $\frac{1}{6}$
- B. $\frac{1}{2}$
- C. $\frac{5}{6}$
- D. $\frac{29}{30}$

6. Probability, 2ADV S1 2019 HSC 6 MC

A game is played by tossing an ordinary 6-sided die and an ordinary coin at the same time. The game is won if the uppermost face of the die shows an even number or the uppermost face of the coin shows a tail (or both).

What is the probability of winning this game?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. $\frac{3}{4}$
- D. 1

7. Probability, 2ADV S1 2014 HSC 10 MC

Three runners compete in a race. The probabilities that the three runners finish the race in under 10 seconds are $\frac{1}{4}$, $\frac{1}{6}$ and $\frac{2}{5}$ respectively.

What is the probability that at least one of the three runners will finish the race in under 10 seconds?

- A. $\frac{1}{60}$
- B. $\frac{37}{60}$
- C. $\frac{3}{8}$
- D. $\frac{5}{8}$

8. Probability, 2ADV S1 2023 MET2 8 MC

A box contains n green balls and m red balls. A ball is selected at random, and its colour is noted. The ball is then replaced in the box.

In 8 such selections, where $n \neq m$, what is the probability that a green ball is selected at least once?

- A. $8\left(\frac{n}{n+m}\right)\left(\frac{m}{n+m}\right)^7$
- B. $1 - \left(\frac{n}{n+m}\right)^8$
- C. $1 - \left(\frac{m}{n+m}\right)^8$
- D. $1 - \left(\frac{n}{n+m}\right)\left(\frac{m}{n+m}\right)^7$

9. Probability, 2ADV S1 2018 HSC 6 MC

A runner has four different pairs of shoes.

If two shoes are selected at random, what is the probability that they will be a matching pair?

- A. $\frac{1}{56}$
- B. $\frac{1}{16}$
- C. $\frac{1}{7}$
- D. $\frac{1}{4}$

10. Probability, 2ADV S1 SM-Bank 6 MC

A box contains n marbles that are identical in every way except colour, of which k marbles are coloured red and the remainder of the marbles are coloured green. Two marbles are drawn randomly from the box.

If the first marble is **not** replaced into the box before the second marble is drawn, then the probability that the two marbles drawn are the same colour is

- A. $\frac{k^2 + (n - k)^2}{n^2}$
- B. $\frac{k^2 + (n - k - 1)^2}{n^2}$
- C. $\frac{2k(n - k - 1)}{n(n - 1)}$
- D. $\frac{k(k - 1) + (n - k)(n - k - 1)}{n(n - 1)}$

11. Probability, 2ADV S1 2022 HSC 9 MC

Liam is playing two games. He is equally likely to win each game. The probability that Liam will win at least one of the games is 80%.

Which of the following is closest to the probability that Liam will win both games?

- A. 31%
- B. 40%
- C. 55%
- D. 64%

12. Probability, 2ADV S1 2004 HSC 1e

A packet contains 12 red, 8 green, 7 yellow and 3 black jellybeans.

One jellybean is selected from the packet at random.

What is the probability that the selected jellybean is red or yellow? *(2 marks)*

13. Probability, 2ADV S1 2014 HSC 12c

A packet of lollies contains 5 red lollies and 14 green lollies. Two lollies are selected at random without replacement.

- i. Draw a tree diagram to show the possible outcomes. Include the probability on each branch. *(2 marks)*
- ii. What is the probability that the two lollies are of different colours? *(1 mark)*

14. Probability, 2ADV S1 2019 HSC 11f

A bag contains 5 green beads and 7 purple beads. Two beads are selected at random, without replacement.

What is the probability that the two beads are the same colour? *(2 marks)*

15. Probability, 2ADV S1 2006 HSC 4c

A chessboard has 32 black squares and 32 white squares. Tanya chooses three different squares at random.

- i. What is the probability that Tanya chooses three white squares? *(2 marks)*
- ii. What is the probability that the three squares Tanya chooses are the same colour?. *(1 mark)*
- iii. What is the probability that the three squares Tanya chooses are not the same colour? *(1 mark)*

16. Probability, 2ADV S1 2007 HSC 4b

Two ordinary dice are rolled. The score is the sum of the numbers on the top faces.

- i. What is the probability that the score is 10? *(2 marks)*
- ii. What is the probability that the score is not 10? *(1 mark)*

17. Probability, 2ADV S1 2012 HSC 13c

Two buckets each contain red marbles and white marbles. Bucket *A* contains 3 red and 2 white marbles. Bucket *B* contains 3 red and 4 white marbles.

Chris randomly chooses one marble from each bucket.

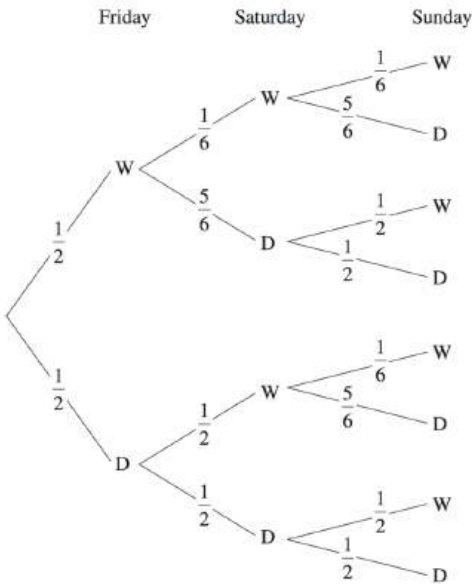
- i. What is the probability that both marbles are red? (1 mark)
- ii. What is the probability that at least one of the marbles is white? (1 mark)
- iii. What is the probability that both marbles are the same colour? (2 marks)

18. Probability, 2ADV S1 2015 HSC 14b

Weather records for a town suggest that:

- if a particular day is wet (*W*), the probability of the next day being dry is $\frac{5}{6}$
- if a particular day is dry (*D*), the probability of the next day being dry is $\frac{1}{2}$.

In a specific week Thursday is dry. The tree diagram shows the possible outcomes for the next three days: Friday, Saturday and Sunday.



- i. Show that the probability of Saturday being dry is $\frac{2}{3}$. (1 mark)
- ii. What is the probability of both Saturday and Sunday being wet? (2 marks)
- iii. What is the probability of at least one of Saturday and Sunday being dry? (1 mark)

19. Probability, 2ADV S1 2010 HSC 4c

There are twelve chocolates in a box. Four of the chocolates have mint centres, four have caramel centres and four have strawberry centres. Ali randomly selects two chocolates and eats them.

- i. What is the probability that the two chocolates have mint centres? (1 mark)
- ii. What is the probability that the two chocolates have the same centre? (1 mark)
- iii. What is the probability that the two chocolates have different centres? (1 mark)

20. Probability, 2ADV S1 2009 HSC 5b

On each working day James parks his car in a parking station which has three levels. He parks his car on a randomly chosen level. He always forgets where he has parked, so when he leaves work he chooses a level at random and searches for his car. If his car is not on that level, he chooses a different level and continues in this way until he finds his car.

- i. What is the probability that his car is on the first level he searches? (1 mark)
- ii. What is the probability that he must search all three levels before he finds his car? (1 mark)
- iii. What is the probability that on every one of the five working days in a week, his car is not on the first level he searches? (1 mark)

21. Probability, 2ADV S1 2019 HSC 15d

The probability that a person chosen at random has red hair is 0.02

- a. Two people are chosen at random.
What is the probability that at least ONE has red hair? (2 marks)
- b. What is the smallest number of people that can be chosen at random so that the probability that at least ONE has red hair is greater than 0.4? (2 marks)

22. Probability, 2ADV S1 2024 HSC 18

In a game, the probability that a particular player scores a goal at each attempt is 0.15.

- a. What is the probability that this player does NOT score a goal in the first two attempts? (1 mark)
- b. Determine the least number of attempts that this player must make so that the probability of scoring at least one goal is greater than 0.8. (2 marks)

23. Probability, 2ADV S1 2004 HSC 6c

In a game, a turn involves rolling two dice, each with faces marked 0, 1, 2, 3, 4 and 5. The score for each turn is calculated by multiplying the two numbers uppermost on the dice.

- i. What is the probability of scoring zero on the first turn? (2 marks)
- ii. What is the probability of scoring 16 or more on the first turn? (1 mark)
- iii. What is the probability that the sum of the scores in the first two turns is less than 45? (2 marks)

24. Probability, 2ADV S1 2005 HSC 5d

A total of 300 tickets are sold in a raffle which has three prizes. There are 100 red, 100 green and 100 blue tickets.

At the drawing of the raffle, winning tickets are NOT replaced before the next draw.

- i. What is the probability that each of the three winning tickets is red? (2 marks)
- ii. What is the probability that at least one of the winning tickets is not red? (1 mark)
- iii. What is the probability that there is one winning ticket of each colour? (2 marks)

25. Probability, 2ADV S1 2007 HSC 9b

A pack of 52 cards consists of four suits with 13 cards in each suit.

- i. One card is drawn from the pack and kept on the table. A second card is drawn and placed beside it on the table. What is the probability that the second card is from a different suit to the first? (1 mark)
- ii. The two cards are replaced and the pack shuffled. Four cards are chosen from the pack and placed side by side on the table. What is the probability that these four cards are all from different suits? (2 marks)

26. Probability, 2ADV S1 2016 HSC 15b

An eight- sided die is marked with numbers 1, 2, ... , 8. A game is played by rolling the die until an 8 appears on the uppermost face. At this point the game ends.

- i. Using a tree diagram, or otherwise, explain why the probability of the game ending before the fourth roll is

$\frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8}$. (2 marks)

- ii. What is the smallest value of *n* for which the probability of the game ending before the *n*th roll is more than $\frac{3}{4}$? (3 marks)

27. Probability, 2ADV S1 2018 HSC 14e

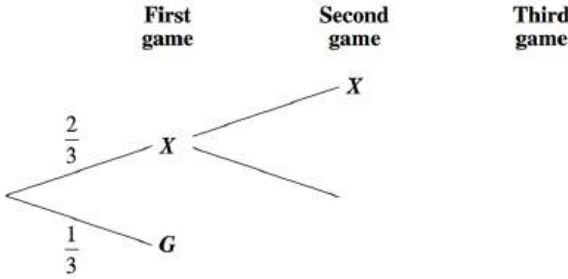
Two machines, *A* and *B*, produce pens. It is known that 10% of the pens produced by machine *A* are faulty and that 5% of the pens produced by machine *B* are faulty.

- i. One pen is chosen at random from each machine.
What is the probability that at least one of the pens is faulty? (1 mark)
- ii. A coin is tossed to select one of the two machines. Two pens are chosen at random from the selected machine.
What is the probability that neither pen is faulty? (2 marks)

28. Probability, 2ADV S1 2008 HSC 7c

Xena and Gabrielle compete in a series of games. The series finishes when one player has won two games. In any game, the probability that Xena wins is $\frac{2}{3}$ and the probability that Gabrielle wins is $\frac{1}{3}$.

Part of the tree diagram for this series of games is shown.



- i. Complete the tree diagram showing the possible outcomes. (1 mark)
- ii. What is the probability that Gabrielle wins the series? (2 marks)
- iii. What is the probability that three games are played in the series? (2 marks)

29. Probability, 2ADV S1 2008 HSC 9a

It is estimated that 85% of students in Australia own a mobile phone.

- i. Two students are selected at random. What is the probability that neither of them owns a mobile phone? (2 marks)
- ii. Based on a recent survey, 20% of the students who own a mobile phone have used their mobile phone during class time. A student is selected at random. What is the probability that the student owns a mobile phone and has used it during class time? (1 mark)

30. Probability, 2ADV S1 2022 HSC 15

In a bag there are 3 six-sided dice. Two of the dice have faces marked 1, 2, 3, 4, 5, 6. The other is a special die with faces marked 1, 2, 3, 5, 5, 5.

One die is randomly selected and tossed.

- a. What is the probability that the die shows a 5? (1 mark)
- b. Given that the die shows a 5, what is the probability that it is the special die? (1 mark)

31. Probability, 2ADV S1 2010 HSC 8b

Two identical biased coins are tossed together, and the outcome is recorded.

After a large number of trials it is observed that the probability that both coins land showing heads is 0.36.

What is the probability that both coins land showing tails? (2 marks)

32. Probability, 2ADV S1 2013 HSC 15d

Pat and Chandra are playing a game. They take turns throwing two dice. The game is won by the first player to throw a double six. Pat starts the game.

- Find the probability that Pat wins the game on the first throw. **(1 mark)**
- What is the probability that Pat wins the game on the first or on the second throw? **(2 marks)**
- Find the probability that Pat eventually wins the game. **(2 marks)**

33. Probability, 2ADV S1 2018 HSC 16b

A game involves rolling two six-sided dice, followed by rolling a third six-sided die. To win the game, the number rolled on the third die must lie between the two numbers rolled previously. For example, if the first two dice show 1 and 4, the game can only be won by rolling a 2 or 3 with the third die.

- What is the probability that a player has no chance of winning before rolling the third die? **(2 marks)**
- What is the probability that a player wins the game? **(2 marks)**

34. Probability, 2ADV S1 2009 HSC 9a

Each week Van and Marie take part in a raffle at their respective workplaces.

The probability that Van wins a prize in his raffle is $\frac{1}{9}$. The probability that Marie wins a prize in her raffle is $\frac{1}{16}$.

What is the probability that, during the next three weeks, at least one of them wins a prize? **(2 marks)**

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Worked Solutions

1. Probability, 2ADV S1 2023 HSC 2 MC

		SPINNER			
		1	2	3	4
DIE	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8
	5	6	7	8	9
	6	7	8	9	10

$$P(\text{score } 7+) = \frac{10}{24} = \frac{5}{12}$$

$\Rightarrow D$

2. Probability, 2ADV S1 2015 HSC 4 MC

$$P(\text{win at soccer}) = \frac{5}{7}$$

$$\therefore P(\text{not win at soccer}) = 1 - \frac{5}{7} = \frac{2}{7}$$

$$P(\text{win at league}) = \frac{2}{3}$$

$$\therefore P(\text{not win at league}) = \frac{1}{3}$$

$$\begin{aligned} \therefore P(\text{not win at both}) &= \frac{2}{7} \times \frac{1}{3} \\ &= \frac{2}{21} \end{aligned}$$

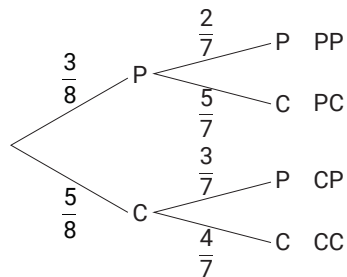
$\Rightarrow A$

3. Probability, 2ADV S1 2019 MET2-N 9 MC

$$\begin{aligned} P(\text{alternate colours}) &= P(RGR) + P(GRG) \\ &= \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} + \frac{2}{5} \cdot \frac{3}{4} \cdot \frac{1}{3} \\ &= \frac{12}{60} + \frac{6}{60} \\ &= \frac{3}{10} \end{aligned}$$

$\Rightarrow B$

4. Probability, 2ADV S1 2021 HSC 6 MC



$$\begin{aligned}
 P(\text{different centres}) &= P(PC) + P(CP) \\
 &= \frac{3}{8} \cdot \frac{5}{7} + \frac{5}{8} \cdot \frac{3}{7} \\
 &= \frac{15}{56} + \frac{15}{56} \\
 &= \frac{15}{28}
 \end{aligned}$$

$\Rightarrow D$

5. Probability, 2ADV S1 2013 HSC 5 MC

$$\begin{aligned}
 P(\text{at least 1 red}) &= 1 - P(\text{none red}) \\
 &= 1 - P(B_1) \times P(B_2) \times P(B_3) \\
 &= 1 - \frac{6}{10} \times \frac{5}{9} \times \frac{4}{8} \\
 &= 1 - \frac{120}{720} \\
 &= 1 - \frac{1}{6} \\
 &= \frac{5}{6}
 \end{aligned}$$

$\Rightarrow C$

6. Probability, 2ADV S1 2019 HSC 6 MC

Game lost only if an odd and a head show.

$$\begin{aligned}
 \therefore P(W) &= 1 - P(\text{odd}) \cdot P(\text{head}) \\
 &= 1 - \frac{3}{6} \cdot \frac{1}{2} \\
 &= \frac{3}{4}
 \end{aligned}$$

$\Rightarrow C$

7. Probability, 2ADV S1 2014 HSC 10 MC

$$P(R_1 < 10 \text{ secs}) = \frac{1}{4} \Rightarrow P(\overline{R_1}) = \frac{3}{4}$$

$$P(R_2 < 10 \text{ secs}) = \frac{1}{6} \Rightarrow P(\overline{R_2}) = \frac{5}{6}$$

$$P(R_3 < 10 \text{ secs}) = \frac{2}{5} \Rightarrow P(\overline{R_3}) = \frac{3}{5}$$

$$\begin{aligned}
 \therefore P(\text{at least 1} < 10 \text{ secs}) \\
 &= 1 - P(\text{all} \geq 10 \text{ secs}) \\
 &= 1 - \frac{3}{4} \times \frac{5}{6} \times \frac{3}{5} \\
 &= 1 - \frac{45}{120} \\
 &= \frac{5}{8}
 \end{aligned}$$

$\Rightarrow D$

♦♦ Mean mark 26%

8. Probability, 2ADV S1 2023 MET2 8 MC

In any single selection:

$$P(\text{green}) = \left(\frac{n}{n+m} \right), \quad P(\text{not green}) = \left(\frac{m}{n+m} \right)$$

Let X = choosing a green ball

$$P(X \geq 1) = 1 - \Pr(X = 0)$$

$$= 1 - \left(\frac{m}{n+m} \right)^8$$

$\Rightarrow C$

♦ Mean mark 49%.

9. Probability, 2ADV S1 2018 HSC 6 MC

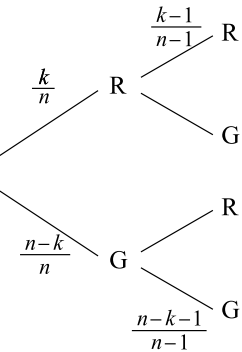
Strategy One:
Choose 1 shoe then find the probability
the next choice is matching.

$$P = 1 \times \frac{1}{7}$$
$$= \frac{1}{7}$$

$$P = \frac{\text{Number of desired outcomes}}{\text{Number of possibilities}}$$
$$= \frac{4}{{}^8C_2}$$
$$= \frac{4}{28}$$
$$= \frac{1}{7}$$
$$\Rightarrow C$$

10. Probability, 2ADV S1 SM-Bank 6 MC

n marbles $\Rightarrow k$ red, $(n-k)$ green



$$\therefore P(\text{same colour}) = \frac{k}{n} \cdot \frac{(k-1)}{(n-1)} + \frac{(n-k)}{n} \cdot \frac{(n-k-1)}{(n-1)}$$
$$= \frac{k(k-1) + (n-k)(n-k-1)}{n(n-1)}$$

$$\Rightarrow D$$

♦♦ Mean mark 31%.

11. Probability, 2ADV S1 2022 HSC 9 MC

$$P(\text{at least 1 W}) = 1 - P(\text{LL}) = 0.8$$

$$P(\text{LL}) = 0.2$$

$$P(\text{L}) = \sqrt{0.2}$$
$$= 0.447$$

$$P(\text{W}) = 1 - 0.447 = 0.553$$

$$P(\text{WW}) = (0.553)^2$$
$$= 0.31$$
$$\Rightarrow A$$

... Mean
marks/comments
here

♦♦♦ Mean mark 27%.

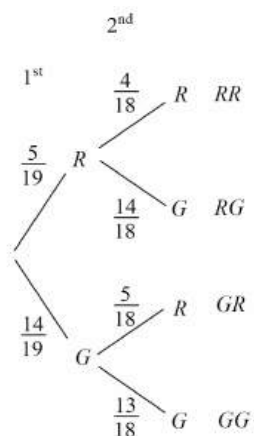
12. Probability, 2ADV S1 2004 HSC 1e

12 R, 8 G, 7 Y, 3 B
Total jellybeans = 30

$$P(\text{R or Y}) = \frac{(\# \text{ Red} + \text{Yellow})}{\text{Total jellybeans}}$$
$$= \frac{12 + 7}{30}$$
$$= \frac{19}{30}$$

13. Probability, 2ADV S1 2014 HSC 12c

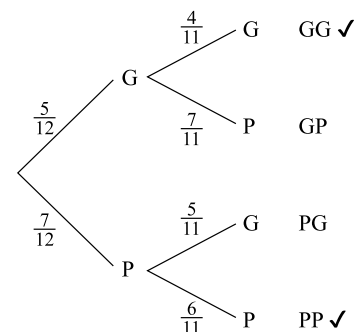
i.



ii. $P(\text{different colours})$

$$\begin{aligned}
 &= P(RG) + P(GR) \\
 &= \frac{5}{19} \times \frac{14}{18} + \frac{14}{19} \times \frac{5}{18} \\
 &= \frac{70}{342} + \frac{70}{342} \\
 &= \frac{140}{342} \\
 &= \frac{70}{171}
 \end{aligned}$$

14. Probability, 2ADV S1 2019 HSC 11f



$$\begin{aligned}
 P(\text{same colour}) &= P(GG) + P(PP) \\
 &= \frac{5}{12} \cdot \frac{4}{11} + \frac{7}{12} \cdot \frac{6}{11} \\
 &= \frac{62}{132} \\
 &= \frac{31}{66}
 \end{aligned}$$

15. Probability, 2ADV S1 2006 HSC 4c

i. $P(WWW) = \frac{32}{64} \times \frac{31}{63} \times \frac{30}{62}$

$$= \frac{5}{42}$$

ii. $P(\text{same colour})$

$$\begin{aligned}
 &= P(WWW) + P(BBB) \\
 &= \frac{5}{42} + \frac{32}{64} \times \frac{31}{63} \times \frac{30}{62} \\
 &= \frac{5}{42} + \frac{5}{42} \\
 &= \frac{5}{21}
 \end{aligned}$$

iii. $P(\text{not all the same colour})$

$$\begin{aligned}
 &= 1 - P(\text{same colour}) \\
 &= 1 - \frac{5}{21} \\
 &= \frac{16}{21}
 \end{aligned}$$

16. Probability, 2ADV S1 2007 HSC 4b

i.

		Die 2					
		1	2	3	4	5	6
Die 1	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

P (score = 10)

= $\frac{3}{36}$
= $\frac{1}{12}$

ii. P (score is not ten)

= 1-P (score is ten)
= $1-\frac{1}{12}$
= $\frac{11}{12}$

17. Probability, 2ADV S1 2012 HSC 13c

i. $P\{(\perp \text{ } hred)\} = P(R_1) \times P(R_2)$

= $\frac{3}{5} \times \frac{3}{7}$
= $\frac{9}{35}$

ii. $P(\text{at least one white}) = 1-P(\text{none white})$

= $1-P(R_1) \times P(R_2)$
= $1-\frac{9}{35}$
= $\frac{26}{35}$

iii. $P(\text{same colour}) = P(R_1R_2) + P(W_1W_2)$

= $\frac{9}{35} + \left(\frac{2}{5} \times \frac{4}{7}\right)$
= $\frac{9}{35} + \frac{8}{35}$
= $\frac{17}{35}$

STRATEGY: When the term "at least" appears in a probability question, it is likely that $1 - P(\text{complement})$ will solve the question more efficiently and with less chance of error, as shown in part (ii).

18. Probability, 2ADV S1 2015 HSC 14b

i. Show $P(\text{Sat dry}) = \frac{2}{3}$

$$\begin{aligned} P(\text{Sat dry}) &= P(W, D) + P(D, D) \\ &= \left(\frac{1}{2} \times \frac{5}{6}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) \\ &= \frac{5}{12} + \frac{1}{4} \\ &= \frac{2}{3} \quad \dots \text{ as required} \end{aligned}$$

ii. $P(\text{Sat and Sun wet})$

$$\begin{aligned} &= P(WWW) + P(DWW) \\ &= \left(\frac{1}{2} \times \frac{1}{6} \times \frac{1}{6}\right) + \left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{6}\right) \\ &= \frac{1}{72} + \frac{1}{24} \\ &= \frac{1}{18} \end{aligned}$$

iii. $P(\text{At least Sat or Sun dry})$

$$\begin{aligned} &= 1 - P(\text{Sat and Sun both wet}) \\ &= 1 - \frac{1}{18} \\ &= \frac{17}{18} \end{aligned}$$

19. Probability, 2ADV S1 2010 HSC 4c

i. $P(2 \text{ mint}) = P(M_1) \times P(M_2)$

$$\begin{aligned} &= \frac{4}{12} \times \frac{3}{11} \\ &= \frac{1}{11} \end{aligned}$$

ii. $P(2 \text{ same}) = P(M_1M_2) + P(C_1C_2) + P(S_1S_2)$

$$\begin{aligned} &= \frac{1}{11} + \left(\frac{4}{12} \times \frac{3}{11}\right) + \left(\frac{4}{12} \times \frac{3}{11}\right) \\ &= \frac{3}{11} \end{aligned}$$

iii. Solution 1

$$\begin{aligned} P(2 \text{ diff}) &= 1 - P(2 \text{ same}) \\ &= 1 - \frac{3}{11} \\ &= \frac{8}{11} \end{aligned}$$

Solution 2

$$\begin{aligned} P(2 \text{ diff}) &= P(M_1, \text{not } M_2) + P(C_1, \text{not } C_2) + P(S_1, \text{not } S_2) \\ &= \left(\frac{4}{12} \times \frac{8}{11}\right) + \left(\frac{4}{12} \times \frac{8}{11}\right) + \left(\frac{4}{12} \times \frac{8}{11}\right) \\ &= \frac{32}{121} + \frac{32}{121} + \frac{32}{121} \\ &= \frac{8}{11} \end{aligned}$$

EXAM TIP: Using $1 - P(2 \text{ things the same})$ is a quicker and easier strategy here.

20. Probability, 2ADV S1 2009 HSC 5b

i. $P(\text{1st chosen}) = \frac{1}{3}$

ii. $P(\text{search 3 levels})$
 $= P(\text{not 1st}) \times P(\text{not 2nd})$
 $= \frac{2}{3} \times \frac{1}{2}$
 $= \frac{1}{3}$

iii. $P(\text{not 1st for 5 days})$
 $= \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$
 $= \frac{32}{243}$

♦♦ Mean marks of 31% and 39% for part (ii) and (iii) respectively.

21. Probability, 2ADV S1 2019 HSC 15d

a. $P(R) = 0.02$

$P(\overline{R}) = 0.98$

$P(\text{At least 1 has red hair})$
 $= 1 - P(\overline{R}, \overline{R})$
 $= 1 - 0.98 \times 0.98$
 $= 0.0396$

b. Find n such that

$1 - 0.98^n > 0.4$
 $0.98^n < 0.6$
 $\ln 0.98^n < \ln 0.6$
 $n \ln 0.98 < \ln 0.6$
 $n > \frac{\ln 0.6}{\ln 0.98}, \quad (\ln 0.98 < 0)$
 $> 25.28 \dots$

$\therefore$ 26 people must be chosen.

♦♦ Mean mark 24%.

22. Probability, 2ADV S1 2024 HSC 18

a. $P(G) = 0.15, \quad P(\tilde{G}) = 0.85$
 $P(\tilde{G}\tilde{G}) = 0.85^2 = 0.7225$

b. 2 attempts: $P(\text{at least 1 goal}) = 1 - P(\tilde{G}\tilde{G}) = 1 - 0.85^2$

3 attempts: $P(\text{at least 1 goal}) = 1 - 0.85^3$

n attempts: $P(\text{at least 1 goal}) = 1 - 0.85^n$

Find n such that:

$1 - 0.85^n > 0.8$

$0.85^n < 0.2$

$n \times \ln(0.85) < \ln(0.2)$

$n > \frac{\ln(0.2)}{\ln(0.85)}$
 $> 9.9 \dots$

$\therefore$ Least $n = 10$

♦ Mean mark (b) 39%.

23. Probability, 2ADV S1 2004 HSC 6c

i.

		Die 1					
		0	1	2	3	4	5
Die 2	0	0	0	0	0	0	0
	1	0	1	2	3	4	5
	2	0	2	4	6	8	10
	3	0	3	6	9	12	15
	4	0	4	8	12	16	20
	5	0	5	10	15	20	25

MARKER'S COMMENT: Students who drew up the table for the sample space were "overwhelmingly" more successful in all parts of this question.

$$\therefore P(0) = \frac{11}{36}$$

$$\text{ii. } P(\geq 16) = \frac{4}{36} = \frac{1}{9}$$

$$\text{iii. } P(\text{Sum} < 45) = 1 - P(\text{Sum} \geq 45)$$

$$\begin{aligned} P(\text{Sum} \geq 45) &= P(20, 25) + P(25, 20) + P(25, 25) \\ &= \left(\frac{2}{36} \times \frac{1}{36} \right) + \left(\frac{2}{36} \times \frac{1}{36} \right) + \left(\frac{1}{36} \times \frac{1}{36} \right) \\ &= \frac{2}{1296} + \frac{2}{1296} + \frac{1}{1296} \\ &= \frac{5}{1296} \end{aligned}$$

$$\begin{aligned} \therefore P(\text{Sum} < 45) &= 1 - \frac{5}{1296} \\ &= \frac{1291}{1296} \end{aligned}$$

24. Probability, 2ADV S1 2005 HSC 5d

$$\begin{aligned} \text{i. } P(RRR) &= \frac{100}{300} \times \frac{99}{299} \times \frac{98}{298} \\ &= \frac{1617}{44\,551} \end{aligned}$$

$$\text{ii. } P(\text{at least 1 winner NOT red})$$

$$\begin{aligned} &= 1 - P(RRR) \\ &= 1 - \frac{1617}{44\,551} \\ &= \frac{42\,934}{44\,551} \end{aligned}$$

$$\text{iii. } \# \text{ Combinations of winning tickets}$$

$$\begin{aligned} &= 3 \times 2 \times 1 \\ &= 6 \end{aligned}$$

$$\therefore P(\text{one winner from each colour})$$

$$\begin{aligned} &= 6 \times \frac{100}{300} \times \frac{100}{299} \times \frac{100}{298} \\ &= 0.22446\dots \\ &= 0.224 \text{ (to 3 d.p.)} \end{aligned}$$

25. Probability, 2ADV S1 2007 HSC 9b

$$\text{i. After 1st card is drawn}$$

$$\# \text{ Cards left from another suit} = 39$$

$$\# \text{ Cards left in pack} = 51$$

$$\therefore P(\text{2nd card from a different suit})$$

$$\begin{aligned} &= \frac{39}{51} \\ &= \frac{13}{17} \end{aligned}$$

$$\text{ii. } P(\text{all 4 cards from different suits})$$

$$\begin{aligned} &= \frac{52}{52} \times \frac{39}{51} \times \frac{26}{50} \times \frac{13}{49} \\ &= \frac{2197}{20\,825} \\ &= 0.1054\dots \\ &= 0.105 \text{ (to 3 d.p.)} \end{aligned}$$

26. Probability, 2ADV S1 2016 HSC 15b

i. $P(\text{game ends before 4th roll})$

$$= P(8) + P(\text{not } 8, 8) + P(\text{not } 8, \text{not } 8, 8)$$
$$= \frac{1}{8} + \frac{7}{8} \cdot \frac{1}{8} + \frac{7}{8} \cdot \frac{7}{8} \cdot \frac{1}{8}$$
$$= \frac{1}{8} + \frac{7}{8} \cdot \frac{1}{8} + \left(\frac{7}{8}\right)^2 \cdot \frac{1}{8} \quad \dots \text{ as required}$$

ii. $\frac{1}{8} + \frac{7}{8} \cdot \frac{1}{8} + \left(\frac{7}{8}\right)^2 \cdot \frac{1}{8} + \dots$

$\Rightarrow$ GP where $a = \frac{1}{8}, \quad r = \frac{7}{8}$

Find n such that $S_{n-1} > \frac{3}{4},$

$$S_{n-1} = \frac{a(1-r^{n-1})}{1-r}$$
$$\frac{3}{4} < \frac{1}{8} \times \frac{\left(1-\left(\frac{7}{8}\right)^{n-1}\right)}{1-\frac{7}{8}}$$
$$\frac{3}{4} < 1-\left(\frac{7}{8}\right)^{n-1}$$
$$\left(\frac{7}{8}\right)^{n-1} < \frac{1}{4}$$
$$(n-1) \cdot \ln \frac{7}{8} < \ln \frac{1}{4}$$
$$n-1 > \frac{\ln \frac{1}{4}}{\ln \frac{7}{8}}$$
$$> 11.38\dots$$

$\therefore n = 12$

♦♦ Mean mark 29%.
ALGEBRA: Note that dividing by $\ln \frac{7}{8}$ reverses the < sign as it is dividing by a negative number.

27. Probability, 2ADV S1 2018 HSC 14e

i. $P(\text{at least 1 faulty}) = 1- P(\text{both faulty})$

$$= 1-0.9 \times 0.95$$
$$= 1-0.855$$
$$= 0.145$$

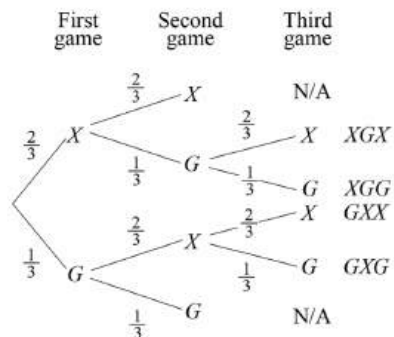
ii. $P(2 \text{ non-faulty pens})$

$$= P(\text{choose A, NF, NF}) + P(\text{choose B, NF, NF})$$
$$= \frac{1}{2} \times 0.9 \times 0.9 + \frac{1}{2} \times 0.95 \times 0.95$$
$$= 0.405 + 0.45125$$
$$= 0.85625$$

♦ Mean mark 48%.

28. Probability, 2ADV S1 2008 HSC 7c

i.



ii. $P(G \text{ wins})$

$$\begin{aligned}
 &= P(XGG) + P(GXG) + P(GG) \\
 &= \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} \\
 &= \frac{2}{27} + \frac{2}{27} + \frac{1}{9} \\
 &= \frac{7}{27}
 \end{aligned}$$

iii. Method 1:

$$\begin{aligned}
 &P(3 \text{ games played}) \\
 &= P(XG) + P(GX) \\
 &= \frac{2}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3} \\
 &= \frac{4}{9}
 \end{aligned}$$

Method 2:

$$\begin{aligned}
 &P(3 \text{ games}) \\
 &= 1 - [P(XX) + P(GG)] \\
 &= 1 - \left[\frac{2}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{3} \right] \\
 &= 1 - \frac{5}{9} \\
 &= \frac{4}{9}
 \end{aligned}$$

MARKER'S COMMENT: A tree diagram with 8 outcomes is incorrect (i.e. no third game is played if 1 player wins the first 2 games). If outcomes cannot occur, do not draw them on a tree diagram.

29. Probability, 2ADV S1 2008 HSC 9a

i. $P(M) = 0.85$

$$P(M^c) = 1 - 0.85 = 0.15$$

$$\begin{aligned}
 \therefore P(M^c, M^c) &= \frac{15}{100} \cdot \frac{15}{100} \\
 &= \frac{225}{10\,000} \\
 &= \frac{9}{400}
 \end{aligned}$$

COMMENT: M^c is syllabus notation for the complement of event M .

ii. $P(\text{owns mobile and used it})$

$$\begin{aligned}
 &= P(M) \times P(\text{used it}) \\
 &= \frac{17}{20} \times \frac{20}{100} \\
 &= \frac{17}{100}
 \end{aligned}$$

30. Probability, 2ADV S1 2022 HSC 15

$$\begin{aligned}
 \text{a. } P(5) &= \frac{1}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{2} \\
 &= \frac{1}{18} + \frac{1}{18} + \frac{1}{6} \\
 &= \frac{5}{18}
 \end{aligned}$$

Mean mark (a) 53%.

♦ Mean mark (b) 40%.

b. $P(\text{special die given a 5 is rolled})$

$$\begin{aligned}
 &= \frac{P(\text{special die}) \cap P(5)}{P(5)} \\
 &= \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{5}{18}} \\
 &= \frac{1}{6} \times \frac{18}{5} \\
 &= \frac{3}{5}
 \end{aligned}$$

31. Probability, 2ADV S1 2010 HSC 8b

$$P(H_1 H_2) = P(H_1) \times P(H_2) \\ = 0.36$$

Since coins are identical:

$$P(H) = \sqrt{0.36} \\ = 0.6$$

$$P(T) = 1 - P(H) \\ = 0.4$$

$$\therefore P(T_1 T_2) = 0.4 \times 0.4 \\ = 0.16$$

♦♦ Mean mark 28%.

NOTE: The most common error was $P(T_1 T_2) = 1 - 0.36 = 0.64$. Ensure you understand why this does **not** apply.

32. Probability, 2ADV S1 2013 HSC 15d

i. P (Pat wins on 1st throw) $= P(W)$

$$P(W) = P(\text{Pat throws 2 sixes}) \\ = \frac{1}{6} \times \frac{1}{6} \\ = \frac{1}{36}$$

ii. Let $P(L) = P(\text{loss for either player on a throw}) = \frac{35}{36}$

$$P(\text{Pat wins on 1st or 2nd throw}) \\ = P(W) + P(LLW) \\ = \frac{1}{36} + \left(\frac{35}{36}\right) \times \left(\frac{35}{36}\right) \times \left(\frac{1}{36}\right) \\ = \frac{2521}{46656} \\ = 0.054 \quad (\text{to 3 d.p.})$$

iii. P (Pat wins eventually)

$$= P(W) + P(LLW) + P(LLLLW) + \dots \\ = \frac{1}{36} + \left(\frac{35}{36}\right)^2 \left(\frac{1}{36}\right) + \left(\frac{35}{36}\right)^4 \left(\frac{1}{36}\right) + \dots$$

$$\Rightarrow \text{GP where } a = \frac{1}{36}, \quad r = \left(\frac{35}{36}\right)^2 = \frac{1225}{1296}$$

Since $|r| < 1$:

$$S_{\infty} = \frac{a}{1-r} \\ = \frac{\frac{1}{36}}{1 - \left(\frac{1225}{1296}\right)} \\ = \frac{1}{36} \times \frac{1296}{71} \\ = \frac{36}{71}$$

$$\therefore \text{Pat's chances to win eventually are } \frac{36}{71}.$$

♦♦ Mean mark 33%

MARKER'S COMMENT: Many students did not account for Chandra having to lose when Pat wins on the **2nd** attempt.

♦♦♦ Mean mark 8%!

COMMENT: Be aware that diminishing probabilities and S_{∞} within the Series and Applications are a natural cross-topic combination.

33. Probability, 2ADV S1 2018 HSC 16b

- i. Construct a sample space of the number of possible winning rolls:

	1	2	3	4	5	6
1	–	–	1	2	3	4
2	–	–	–	1	2	3
3	1	–	–	–	1	2
4	2	1	–	–	–	1
5	3	2	1	–	–	–
6	4	3	2	1	–	–

$$\begin{aligned} P(\text{no chance}) &= \frac{\text{number of pairs with no gap}}{\text{total possibilities}} \\ &= \frac{16}{36} \\ &= \frac{4}{9} \end{aligned}$$

- ii. The sample space in the table shows:

- 8 combinations leave a gap for a single winning number,
- 6 combinations leave a gap for two winning numbers,
- ⋮

$$\begin{aligned} \therefore P(\text{winning}) &= \frac{1}{36} \left[8 \times \frac{1}{6} + 6 \times \frac{2}{6} + 4 \times \frac{3}{6} + 2 \times \frac{4}{6} \right] \\ &= \frac{1}{36} \left(\frac{8}{6} + \frac{12}{6} + \frac{12}{6} + \frac{8}{6} \right) \\ &= \frac{1}{36} \left(\frac{40}{6} \right) \\ &= \frac{5}{27} \end{aligned}$$

♦ Mean mark 40%.
COMMENT: Constructing the full sample space is a critical step here..

♦♦♦ Mean mark 7%.

34. Probability, 2ADV S1 2009 HSC 9a

$$P(\text{Van loses}) = 1 - \frac{1}{9} = \frac{8}{9}$$

$$P(\text{Marie loses}) = 1 - \frac{1}{16} = \frac{15}{16}$$

$$P(\text{both lose}) = \frac{8}{9} \times \frac{15}{16} = \frac{5}{6}$$

$$\begin{aligned} P(\text{At least 1 wins}) &= 1 - P(\text{both lose for 3 weeks}) \\ &= 1 - \left(\frac{5}{6} \right) \left(\frac{5}{6} \right) \left(\frac{5}{6} \right) \\ &= 1 - \frac{125}{216} \\ &= \frac{91}{216} \end{aligned}$$